The Autopoietic Theorem

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Abstract

We show that the autopoietic hierarchy is a formal consequence of three foundational premises within any stable, spatially extended environment. These premises are change (the cosmic ought), finite information capacity (the Bekenstein bound), and stable low-level conditions. From these we derive the three orders of persistence (static, dynamic, and novelty-generating) as a chain of formal consequences. A formal Stack Theoretic interpretation of the Law of Increasing Functional Information shows that persistence under novelty favours weakness maximisation, because this keeps the most compatible futures open. We show that weakness maximisation diverges from simplicity maximisation in stable environments. This bridges a void of unviable intermediate forms and entails self-producing, boundary-maintaining systems. We then invoke the Law of the Stack to show that dynamic persistence at lower levels unlocks higher-level adaptability, creating selection pressure for novelty generation and the preconditions for open-ended evolution. The three orders are thus derivable from axioms rather than identified as empirical regularities. We ground each stage in established models of self-replicating cellular automata, autopoietic protocells, and homeodynamic selves. The result subsumes Assembly Theory's proposed orders of selection while deriving them from weaker assumptions, and reframes the origin of life as a question about mechanism rather than about possibility.

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Introduction

Artificial life studies life as it *could* be (Langton, 1995). Here we prove a theorem of life as it *must* be. Schrödinger asked how life maintains its organisation against entropy (Schrödinger, 1944). Wong et al. (2023) propose a Law of Increasing Functional Information as an answer. Assembly Theory offers a related combinatorial account (Sharma et al., 2023), characterising molecular complexity via the assembly index (the minimum number of joining operations needed to build a molecule from basic parts). All three identify orders of selection as empirical regularities, from static persistence to dynamic persistence and finally novelty generation. We show these orders are derivable from axioms (Bennett, 2025a). The autopoietic hierarchy is not a contingent feature of terrestrial biochemistry. It is a formal consequence of premises that hold in any spatially extended world.

The autopoiesis tradition (Maturana and Varela, 1980; Varela et al., 1974) and the enactivist programme (Thompson, 2007; Paolo, 2005; Froese and Ziemke, 2009; Ciaunica et al., 2023b) characterise living systems as self-producing networks that maintain their own boundaries through organisational closure. Computational models have demonstrated key transitions, including self-replicating CAs (von Neumann, 1966; Langton, 1984; Suzuki and Ikegami, 2006; Sayama, 1999), autopoietic protocells (Suzuki and Ikegami, 2004, 2009; Ono and Ikegami, 2000; Ono, 2005; Beer, 2004; Ruiz-Mirazo et al., 2004), and homeodynamic selves (Ikegami and Suzuki, 2008; Harvey, 2004; Suzuki and Ikegami, 2010). A central theme of this computational programme, particularly in the work of Ikegami and Suzuki, is the emergence of individuality through the dynamic self-maintenance of boundaries (Ikegami and Suzuki, 2008; Beer, 2004). These models demonstrate *that* the transitions are computationally realisable but do not derive *why* they must occur.

Our argument draws on three companion results. A companion paper (Bennett, 2026a) argues that change is the only ontological primitive and that every timeline is already a value judgment about what ought to persist, grounding the cosmic ought without assuming any particular physics. A second companion paper (Bennett, 2026b) and an associated thesis (Bennett, 2025a) use Stack Theory to prove a mathematical analogue of the Law of Increasing Functional Information. They show that persistence favours weakness maximisation (keeping the most compatible futures open) under exchangeable novelty. That result provides the formal engine. The present paper adds the derivation. In stable environments, weakness maximisation diverges from simplicity maximisation, entailing dynamic persistence. The Law of the Stack (Bennett, 2024a) then shows that low-level dynamic persistence is a prerequisite for higher-level adaptability, creating selection pressure for novelty generation. Under a viability prior, weakness maximisation is equivalent to free-energy minimisation (Bennett, 2024a; Friston, 2010), connecting the present account to active inference and interoceptive predictive coding (Seth et al., 2012). Each stage is grounded in computational models that demonstrate its realisability in concrete dynamical systems. Full definitions, proofs, and interpretations are given in the Supplementary Information (SI) (Bennett, 2025b).

Formal Preliminaries

We use a lightweight version of Stack Theory (Bennett, 2025a), which takes an enactivist view of computation (Bennett et al., 2024). The SI gives complete definitions and proofs (Bennett, 2025b). Here we provide just enough to state the results.

We begin by assuming that no matter what universe we are in there is *change*, because without change there is nothing (Whitehead, 1929; Bennett, 2025a). The fact of change implies a set Φ of possible changes. We do not assign changes any ontological content or semantic meaning. Rather, each represents a change or difference in the universe from all the other changes, regardless of whatever the universe happens to be. Hence each $\phi \in \Phi$ is mutually exclusive with all the others, and so we call Φ the *environment* and the members of Φ *states* of the environment. In physics terms you might think of each $\phi \in \Phi$ as a microstate for the whole universe. Because of the mutual exclusivity this equates time with change, so a state is just a point of change and a possible timeline or world is just a sequence of states. A program is just a constraint (a property that can hold in some of those configurations). Continuity is implicit in the fact of infinite states.

A *program* is a set of environmental states (a truth condition). An *embodied language* is a finite set L of *outcomes*, where each outcome $\pi \in L$ is a set of programs (a bundle of constraints the body asserts or imposes on the surrounding environment).

Here we come to our second premise, which is that L is finite. Finiteness follows from the Bekenstein bound (Bekenstein, 1981; Bennett, 2024c). Because outcomes are sets of programs, one can contain another. If $\pi \subseteq y$ (every program in π is also in y) then y is a *completion* of π. The *completion set* of π is

$$\mathrm{Ext}(\pi) := \{y \in L \mid \pi \subseteq y\},$$

and its *weakness* is $w(\pi) := |\mathrm{Ext}(\pi)|$.

Interpretation. Embodiment is where we go from infinite and thus continuous, to finite and thus discontinuous. Spatial extension makes systems finite, which discretises an otherwise continuous world. Weakness refers to weakness of *constraint*. Higher weakness means more degrees of freedom remain. Weakness is the count of how many things the body can still do while doing π (the count of how many futures are left open in the completion set of π). For example, specifying that a human raises one arm is a weaker constraint than specifying that human raises two arms (in the one arm case the other arm, legs, and head can still do many things). Specifying that a human is in a straitjacket imposes a tight constraint and thus low weakness. The completion set is a formal counterpart of Kauffman's adjacent possible (Kauffman, 2000). Crucially, weakness depends on L, not only on π, so it is not reducible to simplicity (Bennett, 2024c).

A *vocabulary* v is a finite set of programs. It generates an embodied language L_v, which is the set of simultaneously satisfiable subsets of v (see SI). This constitutes an *abstraction layer* (Bennett, 2025a, 2024a). An abstraction layer is *stable* when its defining constraints persist across many state transitions. A *stack* is a sequence of layers where each higher layer is induced by the policy adopted below (see SI). This formalises the multiscale competency architecture of biological systems (Levin, 2022).

Interpretation. Molecules form cells, cells form tissues, tissues form organisms. Each is an abstraction layer, and the behaviour at one level constrains what is possible at the next. A stack is a formal model of this hierarchy.

Fix an observed history $\alpha \subseteq L$ (outcomes that have already occurred). The *unobserved region* is $U := L \setminus \alpha$. A policy π is *currently viable* if $\alpha \subseteq \mathrm{Ext}(\pi)$, meaning π has not been falsified. The *unobserved buffer* of a viable π is $B_\pi := \mathrm{Ext}(\pi) \cap U$. Since viability gives $\mathrm{Ext}(\pi) \cap \alpha = \alpha$, we have $|B_\pi| = w(\pi) - |\alpha|$ (SI, Lemma S1). The buffer is how much untested future the policy leaves compatible. For a fixed history, maximising weakness (what previous work labelled w-maxing) is equivalent to maximising the buffer.

Change is fundamental. Without difference, nothing can be distinguished (Whitehead, 1929; Bennett, 2025a). The environment preserves what persists (sets of programs that remain true in long timelines), which formalises a generalised natural selection (Campbell, 1960). We call this the *cosmic ought* (Bennett, 2026a, 2025a)[1].

The Three Orders of Persistence

First order: static persistence

The simplest persistence strategy is structural stability. A system with fewer essential components has fewer ways to break. If each of the $|\pi|$ constraints a body must maintain

[1] Note that while this has a lot in common with Varela's *Calculus of Self Reference* Varela (1975), it diverges significantly from that framework in formalism, in its framing of time and its broader conclusions. These differences are addressed in a companion paper Bennett (2026a).

can be independently disrupted with probability p per step, survival without repair decays as $(1-p)^{|\pi|\,t}$ (Barlow and Proschan, 1975), maximised by minimising $|\pi|$. Previous work named this *simp-maxing* (Bennett, 2026b).

Interpretation. A rock persists because it has little to break. Simp-maxing is the rock strategy. It minimises parts and thus failure modes. The first objects to persist are simple, robust structures like stable atoms and crystals. In Stack Theory these are *static* abstraction layers, pockets of free energy persisting without active maintenance.

Artificial chemistry (Dittrich et al., 2001) illustrates this structural stability but *also* demonstrates an inherent drive toward minimality. The formalism predicts that static replicators should collapse toward minimal forms, and that is just what we observe repeatedly across different computational substrates. In cellular automata, Langton's loops (Langton, 1984) and Suzuki and Ikegami's spatial models (Suzuki and Ikegami, 2006) show that smaller loops replicate faster and eventually dominate the population. Outside the CA paradigm, Fontana and Buss's Algorithmic Chemistry (based on lambda calculus, not spatial dynamics) showed that self-replicating reactions converge to the simplest possible forms such as the identity function (Fontana and Buss, 1994), in the absence of specific constraints that is. Ikegami (1999) found that without adequate external noise, self-reproducing networks of machines and tapes converge to simple autocatalytic core networks. More tellingly, when cells containing such networks were allowed to interact, two lineages emerged. The first was stable cells with fixed core networks. The second was unstable cells whose cores kept reorganising. The unstable (evolvable) lineages could not sustain themselves alone, so they depended *on top of* the population of stable cells as an environment. Once intercellular exchange was removed, the unstable lineages collapsed into stable ones. In each case the system converges on what simp-maxing predicts. Fewer parts, fewer failure modes, and faster copying. Ikegami's two-tier structure also foreshadows the Law of the Stack (Section), in the sense that lower-level stability sustains higher-level exploration.

This convergence is a signature of first-order persistence. As long as the environment is sufficiently uniform that complexity confers no structural advantage, minimising $|\pi|$ dominates. But the simplicity trap is also a ceiling. These systems do not evolve open-ended complexity, because static persistence has no mechanism to cross the void.

The transition. Sayama's Evoloop (Sayama, 1999; Sayama and Nehaniv, 2025), an evolutionary successor to Langton's loops, escapes the simplicity trap by making loops structurally dissolvable. Collisions can destroy any loop, not just small ones. This means fast replication alone no longer guarantees dominance. In a crowded space, offspring are dissolved before they can reproduce. We can move the goalposts so that if offspring reproduce faster, we just make them die faster. This shows survival depends on structural robustness and interaction dynamics rather than copy speed alone. Under these conditions, complexity evolves (Salzberg et al., 2004; Salzberg and Sayama, 2004). In the present terms, universal destructibility undermines the stability condition that lets simp-maxing dominate. The selection landscape shifts toward the regime where weakness and simplicity diverge, and more complex configurations with greater weakness are favoured. Sayama's result is a computational demonstration of the transition the formalism predicts.

Note that these CA loop models are paradigmatically grounded in genetic self-replication, meaning persistence is achieved through structural copying. The autopoietic models of the next section operate on metabolic self-production, meaning persistence through organisational closure. In the limiting case, the genetic-metabolic divide maps onto the static-dynamic distinction. Copying structure is first-order persistence. Maintaining organisation is second-order. Real systems sit on a continuum. For example Sayama's dissolving loops already exhibit interaction dynamics that go beyond pure copying. However the poles of that continuum correspond to the two orders the formalism derives.

Second order: dynamic persistence

Static persistence cannot cope with genuine novelty. A novelty set $S \subseteq U$ represents future demands, which are outcomes that become required after the observed history α. A policy π *survives* S if every newly required outcome is compatible with π, meaning $S \subseteq B_\pi$. You die when the future demands something your embodied commitments already ruled out.

Interpretation. Think of the buffer as breathing room. A policy with a large buffer has made few promises about the future, so there are many ways the future could go that would still be compatible with its commitments. A policy with a small buffer has over-committed, and even a slight surprise could falsify it. Small buffer means fragile.

A formal analogue to the Law of Increasing Functional Information (Bennett, 2026b) shows that a strategy different from simp-maxing dominates under novelty.

Proposition 1 (Survival under novelty favours weakness; restated from Bennett (2026b))**.** *Let π be currently viable. Let P be an exchangeable novelty prior over subsets $S \subseteq U$ (that is, P depends only on $|S|$, not on which elements of U comprise S (de Finetti, 1992)). Then $P(S \subseteq B_\pi)$ is non-decreasing in $|B_\pi|$. Under the uniform prior where every $S \subseteq U$ is equally likely, $\mathbb{P}(survive) = 2^{|B_\pi|-|U|}$.*

Proof sketch. Under the uniform prior, there are $2^{|U|}$ possible novelty sets and $2^{|B_\pi|}$ subsets of the buffer, giving the counting formula directly. For general exchangeable P, condition on $|S| = m$. Given m, the set S is uniform over

$\binom{|U|}{m}$ subsets, so $P(S \subseteq B_\pi \mid |S| = m) = \binom{|B_\pi|}{m} / \binom{|U|}{m}$, which is nondecreasing in $|B_\pi|$ for each m. The unconditional probability is a nonnegative mixture of nondecreasing functions, hence nondecreasing. Full proof in SI. $\square$

Interpretation. If you do not know what the future will demand, you should not overcommit. Under complete ignorance (the uniform prior), every extra compatible unobserved outcome doubles your survival probability. Under any exchangeable prior, survival is monotone in buffer size. More breathing room cannot hurt. Note that there is no complexity term in this expression.

Since $|B_\pi| = w(\pi) - |\alpha|$ for viable policies, larger weakness implies a larger buffer for fixed history size. In unstable environments weakness and simplicity correlate, but in *stable* environments they diverge (Bennett, 2024c, 2025a). This is because a policy π_c with more programs ($|\pi_c| > |\pi_s|$) can have more completions ($w(\pi_c) > w(\pi_s)$) when the structure of L places π_c in a region with many compatible extensions. Self-producing boundary mechanisms are the paradigmatic example. The programs encoding self-production may sit in a part of the language where many continuation strategies remain open.

Interpretation. A self-producing membrane has more parts than a rock, but the structure of the world it inhabits makes those parts open up more possibilities rather than fewer. The rock is simple and limited. The cell is complex but versatile. Which strategy wins depends on whether the environment is stable enough to sustain the complexity.

Information-theoretically, this inversion occurs because autopoietic systems extract environmental correlations that are causally necessary for survival. Kolchinsky and Wolpert (2018) call this *semantic information*. Interventional “shuffling” proves the point. Scramble the correlations between a system and its environment and a passive rock is unaffected, because it carries none. An autopoietic system is destroyed, because its persistence depends on those correlations. The parts of a living system achieve what Biehl et al. (2017) call *complete local integration*. They cannot be decomposed into independent sub-patterns without destroying the whole. Baltieri and Suzuki (2025) review these and related mathematical approaches to distinguishing agents from non-agents. In Stack Theory terms, the programs encoding self-production sit in a region of L where their conjunction achieves high weakness precisely because the parts are integrated. Remove any component and the extension collapses into the void.

This is a novelty-selection analogue of survival of the flattest (Wilke et al., 2001), wherein populations drift toward genotypes with more neutral neighbours (van Nimwegen et al., 1999), and robustness facilitates evolvability by maintaining access to diverse phenotypes (Draghi et al., 2010). Weakness plays the analogous role of maintaining access to diverse futures. The universality of “sloppy” parameter sensitivities in biological models (Gutenkunst et al., 2007; Agosta et al., 2010) provides some empirical support. Most parameters in a biological system can vary freely without affecting function. In the present terms, sloppy parameters sit inside the buffer. They are degrees of freedom left uncommitted. This is weakness in a continuous parameter space.

The interaction between damage and novelty creates a *void of the unviable* (Bennett, 2026b; Solé et al., 2026), composed of those systems complex enough to be fragile, but which do not self-produce and thus do not persist. On one side sit simple static objects. On the other side are complex dynamically persistent systems sustained by organisational closure (Varela et al., 1974; Maturana and Varela, 1980). This resembles the “holey” fitness landscapes of Gavrilets (2004), and the void echoes the unoccupied regions Solé et al. (2026) identify in cognition morphospaces. Organisms that persist at the interface of two environments illustrate how dynamic self-maintenance (here, thermoregulatory homeostasis) is necessary to persist where static robustness alone is insufficient (Watts, 1996). Canadian harbour seals, for example, must continuously manage the thermal boundary between aquatic and terrestrial habitats. Similarly, thermodynamic accounts of life show that persistence of complex structure requires continuous dissipation (England, 2013; Prigogine and Lefever, 1973).

Interpretation. The void is a no-man’s land between rocks and cells. Too complex to survive passively, too simple to self-maintain. Only things on either side persist.

Corollary 2 (Dynamic persistence from novelty selection in stable environments)**.** *In a stable abstraction layer where vocabularies permit complex policies with greater weakness than simple ones, and where autopoietic self-production is a mechanism that achieves this (by maintaining the body in states compatible with many continuations), Proposition 1 implies that persistence filtering enriches survivors with such self-producing systems. The direction of selection toward dynamic persistence is a formal consequence of the Law of Increasing Functional Information operating in stable conditions.*

Interpretation. The corollary does not claim autopoietic systems will arise by any particular mechanism. It claims that *whenever* self-production increases weakness (keeps more futures open), the cosmic ought systematically favours such systems over rocks. The mechanism of origination is empirical. The direction of selection is formal.

Autopoietic protocell models demonstrate this computationally. In Suzuki and Ikegami’s extended autopoiesis model (Suzuki and Ikegami, 2009, 2004), protocells generate and maintain their own boundaries through internal metabolic reactions, which is the hallmark of organisational closure. Different membrane shapes produce different types of self-movement, including chemotaxis. This is dynamic persistence. The system persists not through structural simplicity but through continuous self-production. Yet, to survive unpredictable environments, this autopoietic baseline must be actively defended against severe perturbations.

The homeostatic-to-homeodynamic transition (Ikegami and Suzuki, 2008) adds a further dimension. While a homeostatic system persists by stabilising essential variables within a viable range (often modelled as a fixed point), a homeodynamic system moves beyond this to actively explore new configurations when viability is challenged, via Ashby's ultrastability (Ashby, 1957). The baseline of this stability is illustrated by Harvey's active perception model (Harvey, 2004), where viable conditions are sustained through *rein control*—multiple feedback loops with opposing, one-directional effects producing stability from their collective tension. Within the stack formalism, each drive expands the vocabulary of reachable states, while their opposition constrains the system to a viable region. However, to escape fatal perturbations, the system must transition to homeodynamics. Suzuki and Ikegami's Daisyworld-based agent (Ikegami and Suzuki, 2008) demonstrates how this ultrastability is computationally realised without explicit random parameter searches: internal spatio-temporal chaos (daisy solitons) functions simultaneously as an integrated sensorimotor system, driving itinerant, explorative navigation. In Stack Theory terms, this is not a transition from simp-maxing to w-maxing (the homeostatic system is already complex), but rather a transition from a policy with moderate weakness (fixed-point maintenance via rein control) to one with significantly greater weakness (active, chaotic exploration that opens new continuation strategies). This homeostatic co-regulation can begin between co-embodied organisms (Ciaunica et al., 2021), scaling to elaborate homeodynamic architectures where the underlying information processing is distributed across organismal scales (Ciaunica et al., 2023a; Solé et al., 2024; Levin, 2022; Fields and Levin, 2022).

The computational realisation of this transition is further exemplified by Di Paolo's simulations of homeostatic adaptation (Di Paolo, 2000). In these models, robotic agents are evolved to perform phototaxis (moving toward light). Crucially, the agents are also tasked with keeping their internal "physiological" variables (specifically, the firing rates of neurons in their controllers) within a predefined "viability zone". When these agents are subjected to severe sensorimotor disruptions, such as the inversion of their visual field (swapping left and right sensors), they initially fail to reach the light. This behavioural failure leads to a breach of the viability zone, causing a "loss of homeostasis." Within our Stack Theory framework, this loss of homeostasis represents a collapse of the buffer (B_π). The system's current policy is no longer compatible with environmental demands.

However, Di Paolo's model shows that this breach triggers local structural plasticity. The "need" to restore homeostasis forces the agent to autonomously explore new sensorimotor configurations until it stabilises a new stable relationship with the environment. This is consistent with weakness maximisation. The system does not merely seek a simple solution, but a configuration that restores the greatest number of compatible future states (the "slack") necessary for survival. The "vulnerability" of the system, the fact that it can fail, becomes the very engine that drives it to maintain a wider range of adaptive possibilities.

This principle is being extended into deep learning via Homeostatic Neural Networks (Man et al., 2022). By introducing "vulnerability" and "need" into the learning process, systems can be designed to have "physiological effects" tied to the data they process. For example, in classification tasks, certain objects can be set to alter the learner's internal state (such as learning rate), requiring the agent to selectively "ingest" or "reject" information to maintain functional integrity. In the language of Stack Theory, this active, selective maintenance of the conditions for future action is consistent with weakness maximisation, restoring the "slack" necessary to find new stable states.

Third order: novelty generation

The Law of the Stack (Bennett, 2024a) establishes that higher-level adaptability is bottlenecked by lower-level weakness.

Theorem 3 (The Law of the Stack; Bennett (2024a))**.** *In a stack $\langle v_0, v_1, \dots, v_n \rangle$, let ς_i be the realised policy at layer i. Then the size of the embodied language at the next layer satisfies*

$$\left|L_{v_{i+1}}\right| \leq 2^{|\mathrm{Ext}(\varsigma_i)|} = 2^{w(\varsigma_i)}.$$

Everything expressible at layer $i+1$, including all possible policies and their extensions, is bounded by the weakness of the realised policy at layer i.

Proof sketch. The abstractor maps each completion of ς_i to a truth set, producing v_{i+1}. So $|v_{i+1}| \leq |\mathrm{Ext}(\varsigma_i)|$. The language $L_{v_{i+1}}$ is a subset of $2^{v_{i+1}}$, so $\left|L_{v_{i+1}}\right| \leq 2^{|v_{i+1}|} \leq 2^{|\mathrm{Ext}(\varsigma_i)|}$. Full proof in SI and in Bennett (2024a). □

Interpretation. A narrow lower-layer policy produces a small vocabulary at the next level, capping what behaviours are possible there. To adapt at high levels, a system must retain slack at low levels. This is why delegation of control to lower levels of abstraction facilitates adaptation (Bennett, 2024a).

This echoes Ashby's law of requisite variety (Ashby, 1960). Now apply Proposition 1 at the higher level, and persistence under novelty favours weakness there too. However Theorem 3 caps the weakness achievable at layer $i+1$ (since any policy's extension is a subset of $L_{v_{i+1}}$, its weakness cannot exceed $2^{w(\varsigma_i)}$). Once dynamic persistence provides sufficient lower-level weakness, Proposition 1 at the higher level favours systems that generate novelty. They explore new configurations, producing new behaviours. Policies with large higher-level buffers are pre-adapted to requirements they were never selected for, formalising exaptation (Gould and Vrba, 1982). This is the third order.

Interpretation. Once the biochemistry is taken care of (dynamic persistence), the organism has spare capacity. The Law of the Stack says this spare capacity is what makes behaviour, learning, and communication possible. Natural selection then favours organisms that use it. This is the formal precondition for open-ended evolution. The capacity to generate genuinely new behaviours that were never anticipated by prior selection.

The conversion of lower-level survival requirements into higher-level motivations is formally captured by Homeostatic Reinforcement Learning (HRRL) (Keramati and Gutkin, 2014). Unlike standard RL, which seeks to maximise an external, arbitrary scalar reward, HRRL defines reward as the reduction of internal drive (D). The drive D represents the multi-dimensional distance between the agent's current physiological state (H_t) and an ideal set-point (H^*). The reward r_{t+1} is given by the sequential improvement in this state:

$$r_{t+1} = \beta\,(D_t - D_{t+1}).$$

In this framework, "staying alive" (lower-level persistence) is the direct source of "wanting something" (higher-level motivation). The Law of the Stack is visible here. If the lower-level state is unstable (low weakness), the agent's behaviour is consumed by basic regulation. Only when the lower level achieves a degree of stability can the agent exploit the resulting "slack" to pursue complex environmental goals.

This has been extended using the Nutritional Geometry Framework (Yoshida et al., 2024), where agents must balance multiple competing internal needs (such as proteins and carbohydrates). Simulations show that complex, long-term feeding strategies emerge autonomously from fine-grained motor optimisations, illustrating how lower-level "slack" (the capacity to balance these needs) is a prerequisite for complex multi-objective behaviours.

This demonstrates that high-level behavioural novelty, such as complex search patterns or selective feeding, is not a pre-programmed feature but a consequence of the system's need to maintain a multi-dimensional viability zone. The ability of the "stack" to coordinate these conflicting demands into a single, coherent behavioural policy is what allows for the emergence of sophisticated agency.

The pinnacle of this hierarchy is allostatic control, or "stability through change" (Sterling, 2012). While homeostasis is reactive, allostasis is proactive and predictive. Robotic evaluations of desert-dwelling lizards demonstrate the superiority of allostatic controllers over purely homeostatic ones (Ngo et al., 2022).

An allostatic agent does not wait for its temperature to hit a critical limit. It uses a predictive model of the environment to move toward shade before the internal disruption occurs. Furthermore, allostatic systems can dynamically re-rank competing demands, shifting the priority between energy acquisition, hydration, and safety depending on the context.

From the perspective of Stack Theory, this represents the higher layer of the stack (prediction and planning) rewriting the policy of the lower layer. By proactively adjusting internal set-points, the system expands its overall adaptive range. Recent work by Idei et al. (2025b) has shown that allostatic behaviour can emerge even without fixed set-points, allowing for even more fluid adaptation. This flexibility is consistent with the formal precondition for open-ended evolution. The capacity for a system to generate genuinely new behaviours and cognitive states, ranging from focused attention to exploratory "mind-wandering" (Idei et al., 2025a), all while maintaining the fundamental organisational closure that defines life.

The Autopoietic Theorem

The preceding sections assembled the derivation piece by piece. We now state the combined result.

Theorem 4 (The Autopoietic Theorem)**.** *Let L be a finite embodied language (Bekenstein bound). Assume:*

1. ***Change.*** *The environment produces novelty, modelled by an exchangeable prior P over subsets $S \subseteq U$.*
2. ***Stability.*** *There exist abstraction layers whose defining constraints persist across many state transitions, and whose vocabularies admit policies where self-production increases weakness.*
3. ***Stacking.*** *The system admits a multilayer architecture $\langle v_0, v_1, \ldots\rangle$ where each higher layer is induced by the policy adopted below.*

Then persistence filtering produces the following hierarchy as a chain of formal consequences:

1. ***Static persistence.*** *Among all policies, persistence under damage favours those with minimal $|\pi|$ (simp-maxing). These are the first objects to persist.*
2. ***Dynamic persistence.*** *Under exchangeable novelty (Proposition 1), persistence favours weakness. Given stability (premise 2), self-producing systems have greater weakness than simple systems, so persistence filtering enriches survivors for autopoietic systems (Corollary 2).*
3. ***Novelty generation.*** *The Law of the Stack (Theorem 3) bounds everything expressible at layer $i+1$ by $2^{w(\varsigma_i)}$. Applying Proposition 1 at higher levels, persistence filtering favours systems whose lower-level dynamic persistence provides the slack for higher-level exploration.*

The direction of selection toward each successive order is a formal consequence of the premises. It is not contingent on mechanism, substrate, or terrestrial biochemistry.

Proof. The result is the conjunction of Proposition 1, Corollary 2, and Theorem 3, under the stated premises. Each component is proved above (with more detailed proofs in the SI, and (Bennett, 2024a) in the case of Theorem 3). The conjunction produces the hierarchy because each order creates the preconditions for the next. Static persistence provides stable low-level conditions. Dynamic persistence provides the lower-level weakness that Theorem 3 requires to unlock higher-level expressiveness. Proposition 1 at the higher level then favours novelty generation. □

Interpretation. Given change, finite bodies, and stable ground, the universe has no choice. Rocks come first. Then things that repair themselves. Then things that explore. Each step is forced by the last. The autopoietic hierarchy is not a description of what happened to evolve on Earth. It is a theorem about what must evolve anywhere.

Discussion

Assembly Theory. The three orders derived here are identified empirically by Wong et al. (2023) and Assembly Theory (Sharma et al., 2023). Our derivation adds that they follow from explicit axioms and that the proposed law concerns weakness, not complexity. Whether Assembly Theory's assembly index correlates with weakness in particular systems is an open empirical question.

The void. The predicted void between static and dynamic persistence (Bennett, 2026b; Solé et al., 2026) corresponds to Gavrilets' holey landscapes (Gavrilets, 2004) and Gould's observation that trends toward complexity arise from diffusive spread away from a boundary rather than directional selection (Gould, 1996).

Artificial Life. The Autopoietic Theorem (Theorem 4) suggests that the autopoietic hierarchy is a feature of life as it *must* be, given that these premises hold, not merely as it happens to be on Earth. It is directly relevant to Artificial Life's programme of "life as it could be" (Langton, 1995). The framework also offers a diagnosis of a long-standing puzzle. Artificial self-replicators reliably collapse toward minimal forms (Fontana and Buss, 1994; Ikegami, 1999), and generating open-ended complexity has proven stubbornly difficult. The present account says this is not a modelling failure but a formal consequence. In the absence of stable environmental structure that makes complexity pay, simp-maxing is the only attractor (Section). Systems escape the simplicity trap only when conditions make weakness diverge from simplicity. In the Evoloop, the environment is hostile to all forms equally, but under that universal pressure complexity can pay off in ways it could not before, because fast replication of minimal forms no longer guarantees dominance (Sayama, 1999; Salzberg et al., 2004). More broadly, the derivation identifies the formal preconditions for open-ended evolution. Stable low-level self-maintenance provides the slack for higher-level novelty generation. This connects to the view that intelligence and selfhood are inseparable from embodied self-maintenance (Thompson, 2007; Bennett, 2024b; Ciaunica et al., 2023a).

This synthesis bridges the autopoietic tradition with the quest for open-ended evolution. We reframe homeostasis not as a fixed point, but as the active stabilisation of a viable range that preserves future degrees of freedom. Most importantly, the Law of the Stack identifies lower-layer "slack" (or weakness) as the fundamental functional bottleneck for higher-level novelty. Without the spare capacity provided by lower-level dynamic persistence, the expressive potential of subsequent layers is strictly capped. Thus, lower-level weakness is the formal precondition for the open-ended exploration observed in complex adaptive systems. Ultimately, we hope these principles serve as a theoretical roadmap for designing future ALife models that can successfully realise truly open-ended evolution.

Limitations. The derivation specifies direction of selection, not timescales or mechanisms. Stability is qualitative. The Law of Increasing Functional Information (Bennett, 2026b) is a one-step filtering result. Extending to iterated dynamics with heredity remains open. The exchangeability assumption is analysed in the companion paper's supplementary material, which bounds how far novelty must depart from exchangeability before the buffer-size ordering reverses (Bennett, 2026b). The formal necessity is conditional. Given the premises, the *direction* of selection is necessary, not that life arises with certainty in any finite time. Furthermore, premise 2 of Theorem 4 is a structural condition on the environment, not a claim about the existence of particular systems. The derivation shows that given such structure, the direction of selection is determinate. The question of which physical environments satisfy Premise 2 is empirical; the computational models cited above provide evidence that it is satisfied in realistic conditions.

Conclusion

Langton framed Artificial Life as the study of life as it could be. Here we showed what *must* be. The Autopoietic Theorem shows that in any environment with change, spatial extent, and stable low-level conditions, persistence filtering drives the autopoietic hierarchy as a chain of formal consequences. Rocks, then cells, then minds. Each step is forced by the last. The direction of selection toward life is not one possible outcome among many. It is a theorem of why life must arise.

Abridged Appendix

For complete appendices, see (Bennett, 2025b).

Definition 1 (Environment and states). *An* environment Φ *is a set of mutually exclusive states. Each state $\phi \in \Phi$ is a complete snapshot of reality at a given instant. Because states are mutually exclusive, at most one state obtains at any time.*

Definition 2 (Programs). *A* program p *is a subset of Φ — a set of states. A program p is* true *at state ϕ iff $\phi \in p$. The set of all possible programs is the power set $\mathcal{P} := 2^\Phi$.*

Definition 3 (Vocabulary). *A* vocabulary *is a finite subset $v \subseteq \mathcal{P}$ of programs. The finiteness of v is motivated by the Bekenstein bound Bekenstein (1981): any bounded physical region has finite information capacity, so any embodied system can distinguish only finitely many basic constraints.*

Definition 4 (Outcomes (statements) and embodied language). *Given a vocabulary v, an* outcome *(also called a* statement*) is a subset $\pi \subseteq v$. Asserting π means asserting that every program in π is true simultaneously. The* embodied language *generated by v is*

$$L_v := \{\pi \subseteq v \mid T(\pi) \neq \emptyset\},$$

where $T(\pi) := \bigcap_{p \in \pi} p$ is the set of states at which all programs in π are simultaneously true. The condition $T(\pi) \neq \emptyset$ excludes self-contradictory bundles of constraints. When v is clear from context we write L for L_v.

Definition 5 (Completions, weakness, and complexity). *For an outcome $\pi \in L$:*

- *The* completion set *is* $\mathrm{Ext}(\pi) := \{y \in L \mid \pi \subseteq y\}$.
- *The* weakness *of π is* $w(\pi) := |\mathrm{Ext}(\pi)|$.
- *The* complexity *of π is $|\pi|$ (the number of programs it asserts).*

If $\pi \subseteq y$, then y implies all the constraints π asserts plus possibly more; y is a more specific commitment consistent with π. Weakness counts the number of such more-specific commitments that L contains.

Definition 6 (Observed and unobserved regions, viability, buffer). *Fix a set of outcomes $\alpha \subseteq L$ (a history of observations that have already occurred). Define:*

- Unobserved region*:* $U := L \setminus \alpha$.
- Viability*: a policy π is* currently viable *if $\alpha \subseteq \mathrm{Ext}(\pi)$ (not falsified by any observation).*
- Unobserved buffer*: for viable π, $B_\pi := \mathrm{Ext}(\pi) \cap U$.*

Definition 7 (Task and utility). *A v*-task *τ on vocabulary v specifies a set of inputs $I_\tau \subseteq L_v$, a set of correct outputs $O_\tau \subseteq L_v \setminus I_\tau$, and a set of correct policies $\Pi_\tau := \{\pi \in L_v \mid \mathrm{Ext}(\pi) \cap (L_v \setminus I_\tau) \subseteq O_\tau\}$. A task is satisfiable if $\Pi_\tau \neq \emptyset$. The* utility *of a satisfiable task is*

$$\mathcal{U}(\tau) := \max_{\pi \in \Pi_\tau} w(\pi),$$

the weakness of the weakest (most adaptable) correct policy.

Definition 8 (Abstractor function). *An* abstractor function *f takes a vocabulary v and a statement $l \in L_v$ and returns a sub-vocabulary:*

$$f(v, l) := \{T(o) \mid o \in \mathrm{Ext}(l)\} \subseteq \mathcal{P}.$$

In words: f maps each completion of l to its truth set, and collects these as the next-layer vocabulary.

Definition 9 (Stack). *A* stack *(multilayer architecture) is a sequence $\langle(\tau_0, v_0), (\tau_1, v_1), \dots, (\tau_n, v_n)\rangle$ where:*

- *v_0 is an initial vocabulary with task τ_0.*
- *For each $i < n$: a policy $\varsigma_i \in \Pi_{\tau_i}$ is chosen. Then $v_{i+1} := f(v_i, \varsigma_i)$ and τ_{i+1} is a v_{i+1}-task.*

The policy at layer i determines the vocabulary at layer $i+1$.

Proofs

Lemma: Buffer Size for Viable Policies If π is currently viable (i.e., $\alpha \subseteq \mathrm{Ext}(\pi)$), then $|B_\pi| = w(\pi) - |\alpha|$.

Proof. Since α and $U = L \setminus \alpha$ partition L, the completion set partitions as $\mathrm{Ext}(\pi) = (\mathrm{Ext}(\pi) \cap \alpha) \cup (\mathrm{Ext}(\pi) \cap U)$. These two sets are disjoint because $\alpha \cap U = \emptyset$. The viability assumption $\alpha \subseteq \mathrm{Ext}(\pi)$ gives $\mathrm{Ext}(\pi) \cap \alpha = \alpha$, so $|\mathrm{Ext}(\pi)| = |\alpha| + |B_\pi|$. Rearranging: $|B_\pi| = w(\pi) - |\alpha|$. □

Restatement of Uniform Survival Law. Let U be a finite unobserved region and $B \subseteq U$ a buffer. Let $S \subseteq U$ be drawn uniformly from 2^U. Then $\mathbb{P}(S \subseteq B) = 2^{|B| - |U|}$.

Proof. There are $2^{|U|}$ subsets of U in total and $2^{|B|}$ subsets of B. The event $S \subseteq B$ holds exactly when S is one of the $2^{|B|}$ subsets of B. Under the uniform distribution, $\mathbb{P}(S \subseteq B) = 2^{|B|}/2^{|U|} = 2^{|B| - |U|}$. □

Restatement of Exchangeable Monotonicity. Fix a finite set U and let P be an exchangeable distribution over subsets $S \subseteq U$. Let $B \subseteq U$. Then $P(S \subseteq B)$ is a nondecreasing function of $|B|$.

Proof. Because P is exchangeable, it is fully described by the distribution of $M := |S|$. Condition on $M = m$. Given $M = m$, the set S is uniformly distributed over the $\binom{|U|}{m}$ subsets of U of size m. So

$$P(S \subseteq B \mid M = m) = \begin{cases} \dfrac{\binom{|B|}{m}}{\binom{|U|}{m}} & \text{if } m \leq |B|, \\ 0 & \text{if } m > |B|. \end{cases}$$

For fixed m, the ratio $\binom{|B|}{m}/\binom{|U|}{m}$ is nondecreasing in $|B|$ (adding an element to B can only increase the number of m-subsets contained in B, relative to the fixed total). Taking the expectation over M: a nonnegative weighted sum of nondecreasing functions is nondecreasing. So $P(S \subseteq B)$ is nondecreasing in $|B|$. □

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